24CS2506 – Building Applications using GPT-4

Experiment 1 – Building an AI-Powered E-Commerce Chatbot using Ollama

1. Introduction

Imagine you visit an online shopping website and type:

- Recommend a gaming laptop.
- Is this phone waterproof?
- How can I return my order?
- What do customers think about this product?

Instead of a human customer support executive, an AI chatbot answers these questions instantly. In this experiment, you will build a simplified version of such a chatbot.

2. Understanding the Datasets

Three datasets are used.

Dataset 1 – Products

Contains information about products.

Typical columns:

- Product ID
- Product Name
- Brand
- Category
- Price
- Description

Purpose: This dataset helps the chatbot answer product-related questions.

Dataset 2 – FAQ

Contains frequently asked questions.

Example

Question	Answer
How do I track my order?	You can track your order from My Orders.
How do I cancel my order?	Go to My Orders and click Cancel.

Purpose: Allows the chatbot to answer common customer queries.

Dataset 3 – Reviews

Contains customer opinions.

Example

Product	Review

Product	Review
Laptop	Excellent battery backup
Phone	Camera quality is poor

Purpose: Used for sentiment analysis.

3. What is an AI Chatbot?

An AI chatbot is a computer program capable of understanding user questions and generating appropriate responses. Instead of searching manually, the chatbot identifies the user's intent and provides the most relevant answer.

4. What is a Local LLM?

Large Language Models (LLMs) are AI models capable of understanding and generating human language.

Examples:

- ChatGPT
- Gemini
- Claude
- Qwen
- Llama

In this experiment, **Qwen** runs locally using **Ollama**.

5. What are Embeddings?

Computers cannot understand text directly.

Therefore, sentences are converted into numerical vectors called **embeddings**.

Example

Track my order

↓

[0.21, -0.54, 0.91, ...]

Where is my parcel?

↓

[0.23, -0.51, 0.88, ...]

Although the words differ, the vectors are very similar because both sentences have similar meanings.

6. What is Semantic Search?

Traditional search compares words. Semantic search compares meanings.

Example

User:

Where is my parcel?

FAQ:

How do I track my order?

Although the words differ, both sentences have the same meaning. Semantic search can identify this similarity.

7. What is Cosine Similarity?
Cosine Similarity measures how similar two vectors are.
Value ranges between:
1 → Exactly similar
0 → Completely different
Example

Question	Similarity
Where is my parcel?	0.92
Buy new laptop	0.18

A higher value indicates greater similarity.

8. What is TF-IDF?
TF-IDF stands for:
Term Frequency (TF) - Inverse Document Frequency (IDF)
It identifies the most important words in a document.
Example

Gaming Laptop
↓
gaming
RTX
Intel
SSD

These keywords help recommend similar products.

9. What is Sentiment Analysis?
Sentiment Analysis identifies whether a review expresses a positive, negative, or neutral opinion.
Examples:

Review	Sentiment
Excellent battery life	Positive
Poor display quality	Negative
Average product	Neutral

10. Workflow of the Chatbot



11. Real-Life Applications

- Amazon Customer Support
- Flipkart Assistant
- Myntra Shopping Assistant
- Banking Chatbots
- Airline Booking Systems
- Hospital Appointment Systems